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Supporting Information

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Molecular Insights into the Regulatable Interfacial Property and Flow Behavior of Confined Ionic Liquids in Graphene Nanochannels

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Supporting Information

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Supporting Figures and Captions

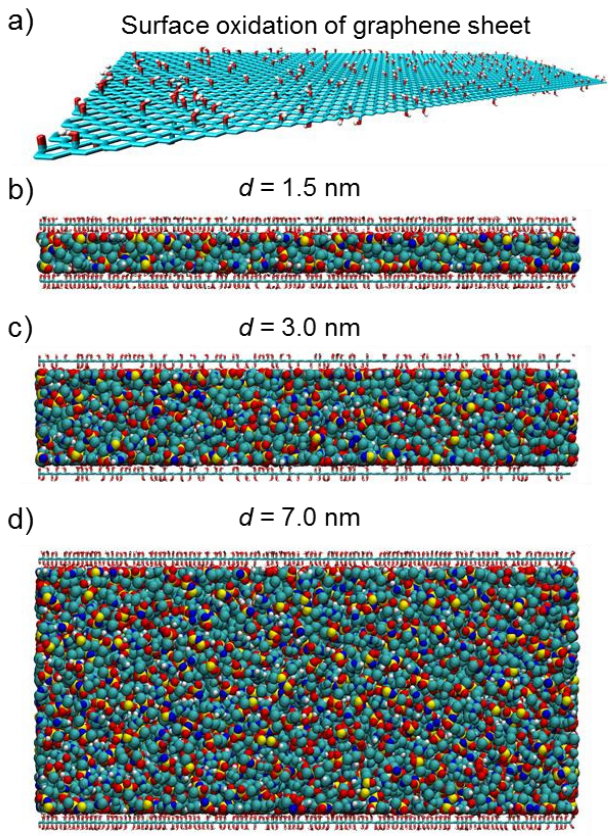


Figure S1. a) The atomic structure of graphene oxide sheet, where the hydroxyl functional groups are randomly distributed on the sheet. b-d) The atomic structure of confined IL inside graphene nanochannel with c_{OH} = 15%.

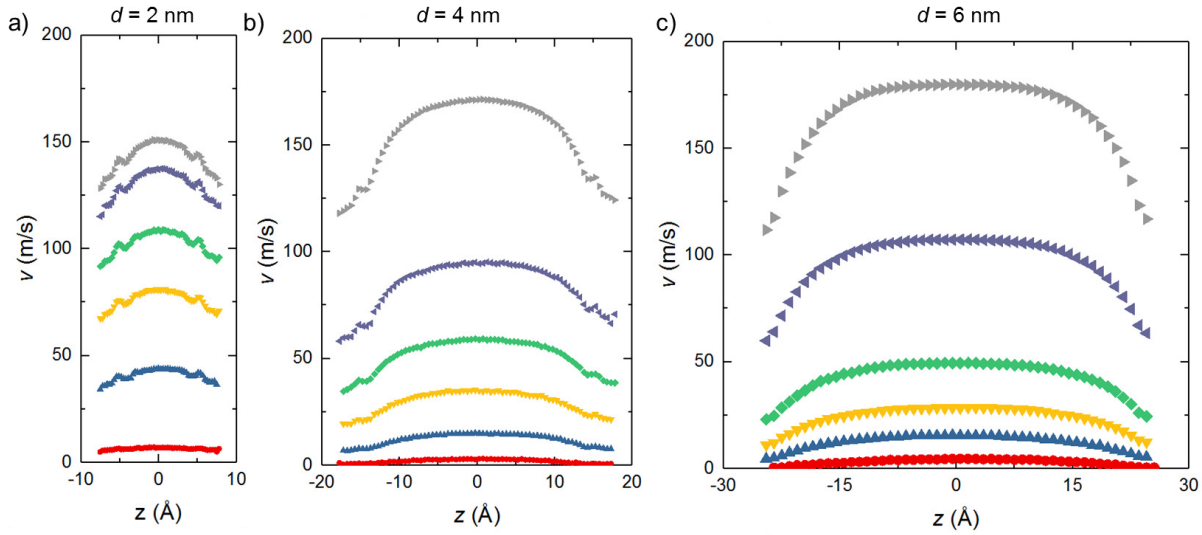


Figure S2. a-c) The velocity profile of IL inside graphene nanochannel with c_{OH} = 15% for d = 2, 4 and 6 nm, showing the typical Poiseuille flow and partial plunger flow for different d and pressure gradient.

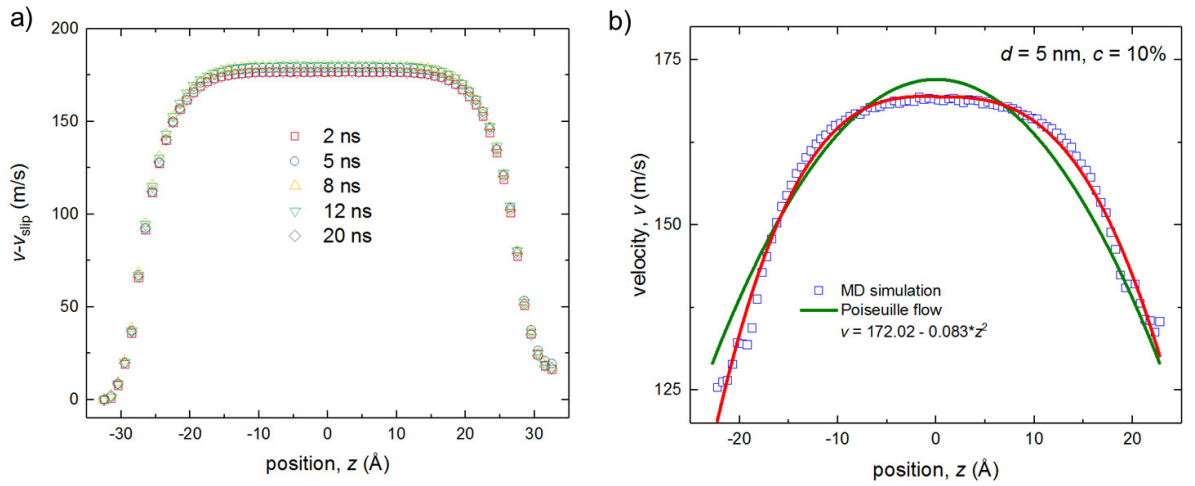


Figure S3. a) The velocity profile of IL inside graphene nanochannel of d = 7 nm with c_{OH} = 15% at a different simulated time, showing that 2 ns is long enough to obtain the convergent velocity. b) The velocity fitting using Poiseuille and partial plunger flow, indicating that the velocity cannot be described well using the Poiseuille flow.

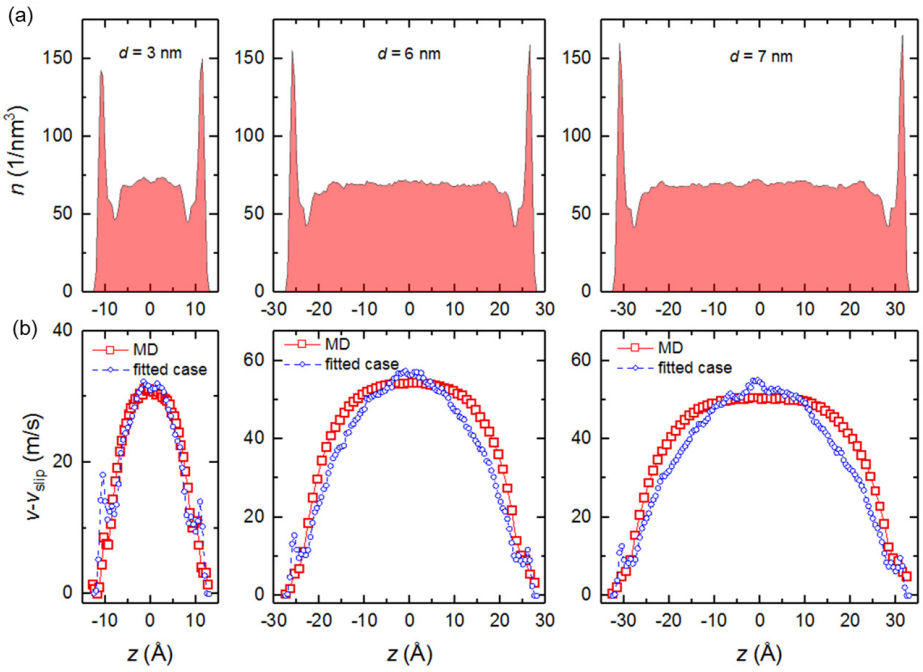


Figure S4. a) The number density of ILs (n) as a function of the position for ILs inside GO nanochannel (c_{OH} = 15% and d = 3, 6, 7 nm). b) The velocity profile from NEMD simulations and fitted value according to the theoretical model $v(z) = -\frac{nF^e}{2\eta}(z^2 - \frac{d^2}{4}) + v_{slip}$ (MRS Bulletin, 2017, 42(4):283-288), showing that the non-linear density distribution is not the direct reason of the partial plunger flow of ILs inside GO nanochannel. The difference between theoretical value and MD results, especially the velocity peak near the wall, should result from the different viscosity of ILs at a different position in the nanochannel.

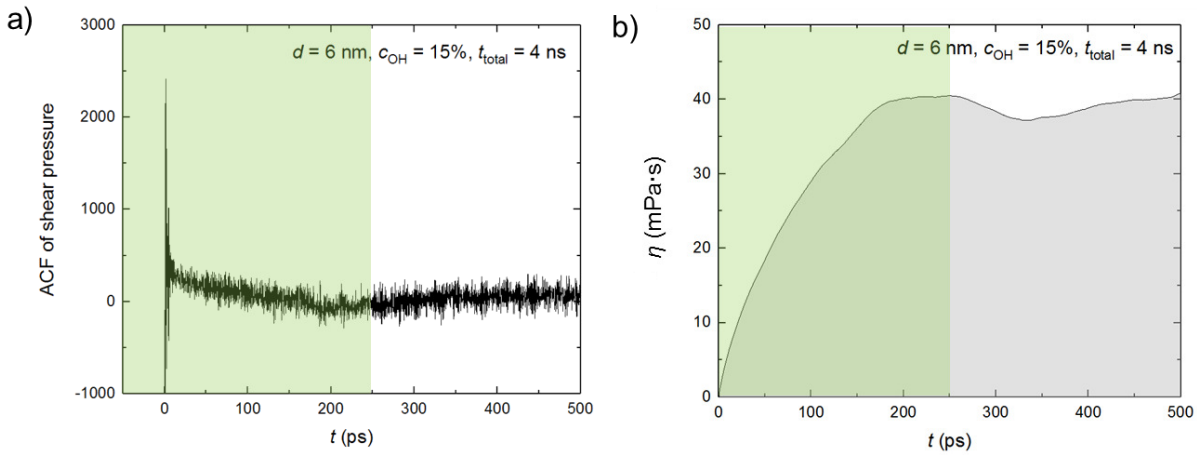


Figure S5. a-b)The autocorrelation function (ACF) of shear pressure and the integral viscosity of IL in graphene nanochannel with d = 6 nm and c_{OH} = 15%, indicating that the viscosity will up to convergence at ~250 ps. In the present work, at least 10 independent 2 ns long simulation were performed to obtain the average viscosity for confined ILs.

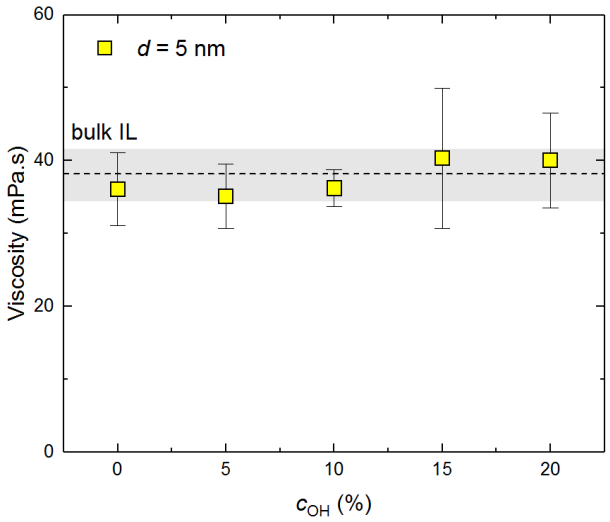


Figure S6. The viscosity for IL confined inside nanochannel with different degree of surface oxidation (d = 5 nm), showing that the viscosity has a weak relation with the surface chemistry. Such effect can be enhanced when IL inside small nanochannel due to the higher fraction of interface region for small nanochannel.

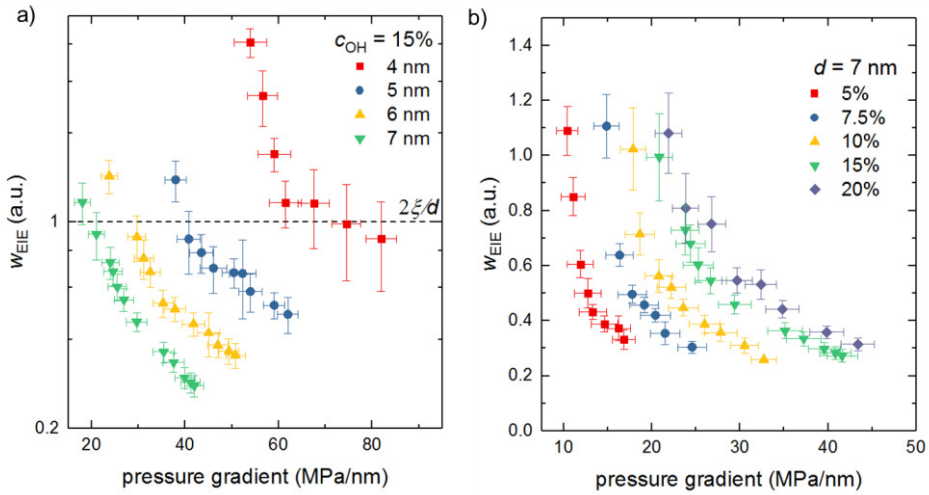


Figure S7. a-b) The effective influence extent of nanochannel, w_{EIE}, on the confined IL as a function of the pressure gradient for different width and surface oxidation degree.

Table S1. The in-plain self-diffusive coefficient of cation and anion of IL inside graphene nanochannel with different d. Units: 10^{-11} m^2/s

d (nm)	1.5	2	3	4	5	bulk
$[Emim]^+$	0.34±0.001	0.95±0.002	2.60±0.004	4.36±0.003	5.09±0.005	6.8±0.007
$[TFSI]^-$	0.29±0.001	0.57±0.001	1.23±0.003	2.12±0.004	2.52±0.006	4.11±0.005

Table S2. The viscosities of bulk liquids obtained from MD simulations at T = 300 K and experimental measurements at ambient conditions.

Liquid	Acetone	Hexane	Toluene	Ethanol	Ethylene Glycol
MD	0.254	0.366	0.501	1.016	10.666
Exp.	0.260	0.326	0.564	1.074	15.977
Liquid	**Butanol**	**Water**	**IL1: $[Emim]^+[TFSI]^-$**	**IL2: $[Bmim]^+[BF_4]^-$**	
MD	1.616	0.729	38.21	108.94	
Exp.	2.300	0.850	32.6	110.308	

Notes: The data for acetone, hexane, toluene, ethanol, ethylene glycol, butanol, and water are from Jiao et al., ACS Appl. Mater. Interf., 2018, DOI: 10.1021/acsami.8b12871. The MD results for ILs are from our work and experimental results are from Aranowski et al., J. Mol. Liq. 2016, 221, 541-546 and Wang et al., Green Chem., 2003, 5, 618-622.

Table S3. The intercalated pairs of IL in the graphene nanochannel with different d and the average mass density.

d (nm)	1.5	2	3	4	5	6	7
N_{IL} (pair)	214	297	500	689	882	1075	1267
ρ_{ave} (g/cm^3)	1.520	1.442	1.492	1.481	1.484	1.487	1.486